

基于 CentOS 7 快速搭建 LAMP 环境

试验前准备

1. 修改 yum 源为阿里云

```
wget -O /etc/yum.repos.d/CentOS-Base.repo https://mirrors.aliyun.com/repo/Centos-7.repo
yum clean all && yum makecache 生成缓存
```

2. 关闭防火墙

```
systemctl stop firewalld //关闭防火墙;
systemctl disable firewalld //禁止防火墙开机启动;
systemctl status firewalld //查看防火墙状态;
```

```
[root@localhost ~]# systemctl status firewalld
● firewalld.service - firewalld - dynamic firewall daemon
   Loaded: loaded (/usr/lib/systemd/system/firewalld.service; disabled; vendor preset: enabled)
   Active: inactive (dead)
     Docs: man:firewalld(1)

8月 01 16:20:49 localhost.localdomain systemd[1]: Starting firewalld - dynamic firewall daemon...
8月 01 16:20:50 localhost.localdomain systemd[1]: Started firewalld - dynamic firewall daemon.
8月 01 17:27:54 localhost.localdomain systemd[1]: Stopping firewalld - dynamic firewall daemon...
8月 01 17:27:55 localhost.localdomain systemd[1]: Stopped firewalld - dynamic firewall daemon.
[root@localhost ~]#
```

3. 关闭 selinux

```
vi /etc/selinux/config //编辑 selinux 配置文件;
```

按 i 键进入插入模式，修改为 SELINUX=disabled，按 ESC 键退回到命令模式，输入:wq 保存并退出;

```
# This file controls the state of SELinux on the system.
# SELINUX= can take one of these three values:
#   enforcing - SELinux security policy is enforced.
#   permissive - SELinux prints warnings instead of enforcing.
#   disabled - No SELinux policy is loaded.
SELINUX=disabled
# SELINUXTYPE= can take one of three values:
#   targeted - Targeted processes are protected,
#   minimum - Modification of targeted policy. Only selected processes are protected.
#   mls - Multi Level Security protection.
SELINUXTYPE=targeted
```

```
reboot //重新启动系统;
```

```
getenforce //查看 selinux 状态，如果是 disabled 则表示成功;
```

```
[root@localhost student]# getenforce
Disabled
```

一. 安装 Apache 服务

1. 登录系统

2. 执行如下命令，安装 Apache 服务及其扩展包。

```
yum -y install httpd httpd-manual mod_ssl mod_perl mod_auth_mysql
```

返回类似如下图结果则表示安装成功。

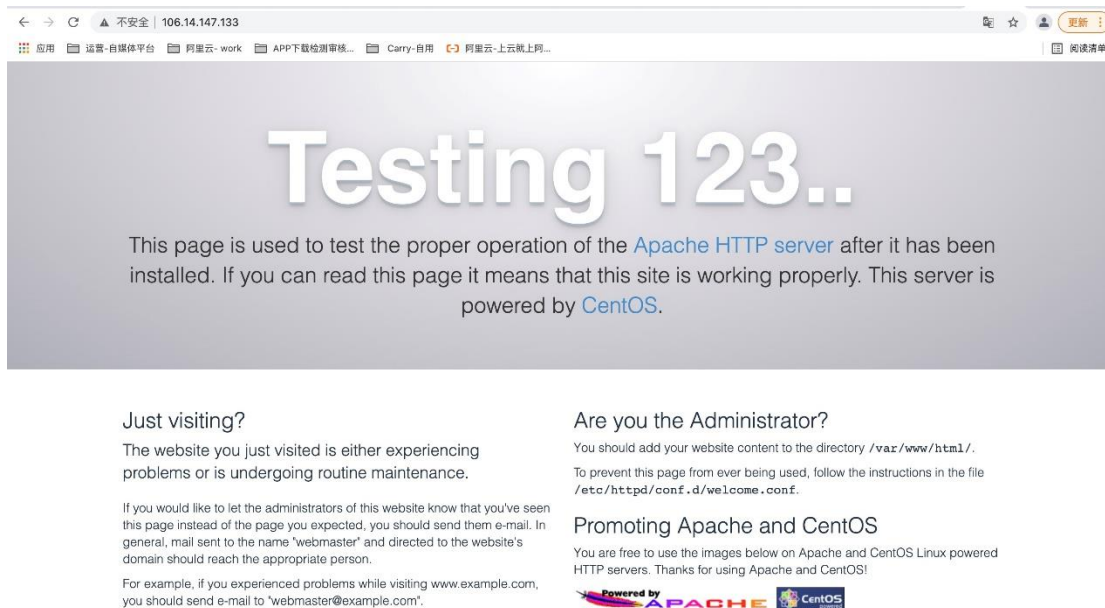
```
Dependency Installed:
apr.x86_64 0:1.4.8-7.el7
glibc-devel.x86_64 0:2.17-324.el7_9
kernel-headers.x86_64 0:3.10.0-1160.36.2.el7
perl-BSD-Resource.x86_64 0:1.29.0-1.el7
perl-ExtUtils-MakeMaker.noarch 0:6.60-3.el7
perl-Linux-Pid.x86_64 0:0.04-18.el7
pyparsing.noarch 0:1.5.6-9.el7
apr-util.x86_64 0:1.5.2-6.el7
glibc-headers.x86_64 0:2.17-324.el7_9
libdb-devel.x86_64 0:5.3.21-25.el7
perl-Data-Dumper.x86_64 0:2.145-3.el7
perl-ExtUtils-Manifest.noarch 0:1.61-244.el7
perl-Test-Harness.noarch 0:3.28-3.el7
systemtap-sdt-devel.x86_64 0:4.0-13.el7
gdbm-devel.x86_64 0:1.10-8.el7
httpd-tools.x86_64 0:2.4.6-97.el7.centos
mailcap.noarch 0:2.1.41-2.el7
perl-ExtUtils-Install.noarch 0:1.58-299.el7_9
perl-ExtUtils-ParseXS.noarch 1:3.18-3.el7
perl-devel.x86_64 4:5.16.3-299.el7_9
Dependency Updated:
glibc.x86_64 0:2.17-324.el7_9
glibc-common.x86_64 0:2.17-324.el7_9
nscd.x86_64 0:2.17-324.el7_9
Complete!
```

3. 执行如下命令，启动 Apache 服务。

Systemctl enable httpd.service
systemctl start httpd.service

4. 在本地电脑的浏览器的址栏中，输入 IP 登录地址，并按 Enter 键。

若返回页面如下图所示，说明 Apache 服务启动成功。



二 . 安装并配置 MySQL

MySQL 是最流行的关系型数据库管理系统，在 WEB 应用方面 MySQL 是最好的 RDBMS(Relational Database Management System：关系数据库管理系统)应用软件之一。

1. 执行以下命令，下载并安装 MySQL 官方的 Yum Repository。

```
wget http://dev.mysql.com/get/mysql57-community-release-el7-10.noarch.rpm  
yum -y install mysql57-community-release-el7-10.noarch.rpm  
yum -y install mysql-community-server --nogpgcheck
```

2. 运行以下命令查看 MySQL 版本号。

```
mysql -V
```

返回结果如下所示，表示 MySQL 安装成功。

```
[root@localhost ~]# mysql -V  
mysql Ver 14.14 Distrib 5.7.35, for Linux (x86_64) using EditLine wrapper
```

3. 执行以下命令，启动 MySQL 数据库。

```
systemctl start mysqld.service
```

4. 执行以下命令，查看 MySQL 初始密码。

```
grep "password" /var/log/mysqld.log
```

```
[root@localhost ~]# grep "password" /var/log/mysqld.log  
2021-08-13T06:27:17.453175Z 1 [Note] A temporary password is generated for root@localhost: hu...FN
```

5. 执行以下命令，登录数据库

```
mysql -uroot -p
```

```
[root@izuf652 ~]# mysql -uroot -p
Enter password:
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 2
Server version: 5.7.35

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owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.
```

6. 输入 MySQL 的初始密码。

说明：在输入密码时，界面不会显示密码。

```
2023-03-16T02:13:27.098676Z 1 [Note] A temporary password is generated for root@localhost: S=e1MDj:)rrg
[root@izuf652 ~]# mysql -uroot -p
Enter password:
```

7. 执行以下命令，修改 MySQL 默认密码。

set global validate_password_policy=0; #修改密码安全策略为低（只校验密码长度，至少 8 位）。

ALTER USER 'root'@'localhost' IDENTIFIED BY '12345678';

```
mysql> set global validate_password_policy=0;
Query OK, 0 rows affected (0.00 sec)

mysql> ALTER USER 'root'@'localhost' IDENTIFIED BY '12345678';
Query OK, 0 rows affected (0.00 sec)
```

8. 执行以下命令，授予 root 用户远程管理权限。

GRANT ALL PRIVILEGES ON *.* TO 'root'@'%' IDENTIFIED BY '12345678';

```
mysql> GRANT ALL PRIVILEGES ON *.* TO 'root'@'%' IDENTIFIED BY '12345678';
Query OK, 0 rows affected, 1 warning (0.00 sec)
```

9. 输入 exit 退出 数据库。

```
mysql> exit
Bye
```

三 . 安装 PHP

PHP（PHP：Hypertext Preprocessor 递归缩写）中文名字是：“超文本预处理器”，是一种广泛使用的通用开源脚本语言，适合于 Web 网站开发，它可以嵌入 HTML 中。编程范型是面向对象、命令式编程的。

1. 执行以下命令，安装 PHP 环境。

```
yum -y install php php-mysql gd php-gd gd-devel php-xml php-common php-mbstring php-ldap php-pear
php-xmlrpc php-imap
```

```

Dependency Installed:
dejavu-fonts-common.noarch 0:2.33-6.el7      dejavu-sans-fonts.noarch 0:2.33-6.el7      expat-devel.x86_64 0:2.1.0-12.el7
fontconfig.x86_64 0:2.13.0-4.3.el7          fontconfig-devel.x86_64 0:2.13.0-4.3.el7    fontpackages-filesystem.noarch 0:1.44-8.el7
freetype-devel.x86_64 0:2.8-14.el7_9.1      libICE.x86_64 0:1.0.9-9.el7               libSM.x86_64 0:1.2.2-2.el7
libX11.x86_64 0:1.6.7-3.el7_9              libX11-common.noarch 0:1.6.7-3.el7_9        libX11-devel.x86_64 0:1.6.7-3.el7_9
libXau.x86_64 0:1.0.8-2.1.el7              libXau-devel.x86_64 0:1.0.8-2.1.el7        libXext.x86_64 0:1.3.3-3.el7
libXpm.x86_64 0:3.5.12-1.el7               libXpm-devel.x86_64 0:3.5.12-1.el7        libXt.x86_64 0:1.1.5-3.el7
libc-client.x86_64 0:2007f-16.el7          libjpeg-turbo.x86_64 0:1.2.90-8.el7        libjpeg-turbo-devel.x86_64 0:1.2.90-8.el7
libpng-devel.x86_64 2:1.5.13-8.el7          libuuid-devel.x86_64 0:2.23.2-65.el7_9.1    libxcb.x86_64 0:1.13-1.el7
libxcb-devel.x86_64 0:1.13-1.el7           libxslt.x86_64 0:1.1.28-6.el7             libzip.x86_64 0:0.10-1-8.el7
php-cli.x86_64 0:5.4.16-48.el7             php-pdo.x86_64 0:5.4.16-48.el7            php-process.x86_64 0:5.4.16-48.el7
t1lib.x86_64 0:5.1.2-14.el7               xorg-x11-proto-devel.noarch 0:2018.4-1.el7  zlib-devel.x86_64 0:1.2.7-19.el7_9

Dependency Updated:
expat.x86_64 0:2.1.0-12.el7                freetype.x86_64 0:2.8-14.el7_9.1          libblkid.x86_64 0:2.23.2-65.el7_9.1      libmount.x86_64 0:2.23.2-65.el7_9
libpng.x86_64 2:1.5.13-8.el7               libsmartcols.x86_64 0:2.23.2-65.el7_9.1    libuuid.x86_64 0:2.23.2-65.el7_9.1        util-linux.x86_64 0:2.23.2-65.el7_9
zlib.x86_64 0:1.2.7-19.el7_9

Complete!

```

2. 执行以下命令创建 PHP 测试页面。

```
echo "<?php phpinfo(); ?>" > /var/www/html/phpinfo.php
```

```
[root@i7-4172 ~]# echo "<?php phpinfo(); ?>" > /var/www/html/phpinfo.php
```

3. 执行以下命令，重启 Apache 服务。

```
systemctl restart httpd
```

```
[root@i7-4172 ~]# systemctl restart httpd
```

4. 在本地浏览器的址栏中，访问 <http://IP 地址/phpinfo.php>，显示如下页面表示 PHP 语言环境安装成功。

System	Linux iZuf6b6ypggfjmOp9rtlkZ 3.10.0-1062.18.1.el7.x86_64 #1 SMP Tue Mar 17 23:49:17 UTC 2020 x86_64
Build Date	Apr 1 2020 04:08:16
Server API	Apache 2.0 Handler
Virtual Directory Support	disabled
Configuration File (php.ini) Path	/etc
Loaded Configuration File	/etc/php.ini
Scan this dir for additional .ini files	/etc/php.d
Additional .ini files parsed	/etc/php.d/curl.ini, /etc/php.d/dom.ini, /etc/php.d/fileinfo.ini, /etc/php.d/gd.ini, /etc/php.d/imap.ini, /etc/php.d/json.ini, /etc/php.d/ldap.ini, /etc/php.d/mbstring.ini, /etc/php.d/mysql.ini, /etc/php.d/mysqli.ini, /etc/php.d/pdo.ini, /etc/php.d/pdo_mysql.ini, /etc/php.d/pdo_sqlite.ini, /etc/php.d/phar.ini, /etc/php.d/posix.ini, /etc/php.d/sqlite3.ini, /etc/php.d/sysvmsg.ini, /etc/php.d/sysvsem.ini, /etc/php.d/sysvshm.ini, /etc/php.d/wddx.ini, /etc/php.d/xmlreader.ini, /etc/php.d/xmlrpc.ini, /etc/php.d/xmlwriter.ini, /etc/php.d/xsl.ini, /etc/php.d/zip.ini
PHP API	20100412
PHP Extension	20100525
Zend Extension	220100525
Zend Extension Build	API220100525,NTS
PHP Extension Build	API20100525,NTS
Debug Build	no
Thread Safety	disabled
Zend Signal Handling	disabled
Zend Memory Manager	enabled
Zend Multibyte Support	provided by mbstring
IPv6 Support	enabled
DTrace Support	disabled
Registered PHP Streams	https, ftps, compress.zlib, compress.bzip2, php, file, glob, data, http, ftp, phar, zip
Registered Stream Socket Transports	tcp, udp, unix, udg, ssl, sslv3, tls
Registered Stream Filters	zlib.*, bzip2.*, convert.iconv.*, string.rot13, string.toupper, string.tolower, string.strip_tags, convert.*, consumed, dechunk

四. 安装 phpMyAdmin

phpMyAdmin 是一个 MySQL 数据库管理工具，通过 Web 接口管理数据库方便快捷。

1. 执行以下命令，创建 phpMyAdmin 数据存放目录。

```
mkdir -p /var/www/html/phpmyadmin
```

```
[root@iZuf6b6ypggfjmOp9rtlkZ ~]# mkdir -p /var/www/html/phpmyadmin
```

2. 执行以下命令，下载 phpMyAdmin 压缩包。

wget --no-check-certificate https://labfileapp.oss-cn-hangzhou.aliyuncs.com/phpMyAdmin-4.0.10.20-all-languages.zip

```
[root@iz0000000000 ~]# wget https://files.phpmyadmin.net/phpMyAdmin/4.0.10.20/phpMyAdmin-4.0.10.20-all-languages.zip
--2021-08-13 15:38:54-- https://files.phpmyadmin.net/phpMyAdmin/4.0.10.20/phpMyAdmin-4.0.10.20-all-languages.zip
Resolving files.phpmyadmin.net (files.phpmyadmin.net)... 89.187.160.8, 2a02:6ea0:d300::7
Connecting to files.phpmyadmin.net (files.phpmyadmin.net)[89.187.160.8]:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 7457007 (7.1M) [application/zip]
Saving to: 'phpMyAdmin-4.0.10.20-all-languages.zip'

100%[----->] 7,457,007  1.58MB/s  in 4
2021-08-13 15:39:07 (1.49 MB/s) - 'phpMyAdmin-4.0.10.20-all-languages.zip' saved [7457007/7457007]
```

3. 执行以下命令，安装 unzip 并解压 phpMyAdmin 压缩包。

```
yum install -y unzip
```

```
unzip phpMyAdmin-4.0.10.20-all-languages.zip
```

4. 执行以下命令，移动 phpMyAdmin 文件到数据存放目录。

```
mv phpMyAdmin-4.0.10.20-all-languages/* /var/www/html/phpmyadmin
```

5. 在本地浏览器的址栏中，输入 http:// IP 地址/phpmyadmin，访问 phpMyAdmin。

返回页面如下图所示，说明 phpMyAdmin 安装成功。



6. 在 phpMyAdmin 登录页面，依次输入 MySQL 的用户名和密码，单击执行。



The image shows the phpMyAdmin login interface. At the top, there is a logo and the text "欢迎使用 phpMyAdmin". Below this, there is a language selection dropdown set to "中文 - Chinese simplified". The main section is the login form, which includes fields for "用户名:" (Username) and "密码:" (Password). The username field contains "root" and the password field contains a series of dots. A red box highlights the username field, and another red box highlights the password field. A third red box highlights the "执行" (Execute) button. A warning message states: "此连接不安全。在此页面输入的登录信息可以被窃取。详细了解" (This connection is not secure. Login information entered on this page can be intercepted. Learn more).

返回页面如下图所示，表示 MySQL 连接成功。



The image shows the phpMyAdmin main interface after a successful MySQL connection. The browser address bar shows the URL: "/phpmyadmin/index.php?token=3a394ce500d4d827214f85fd02d712b3#PMAURL-0:index.php?db=&". The interface is in Chinese. The left sidebar shows a list of databases: "information_schema", "mysql", "performance_schema", and "sys". The main content area is divided into several sections: "常规设置" (General Settings) with a "修改密码" (Change Password) button and a "服务器连接排序规则" (Server Connection Collation) dropdown; "外观设置" (Appearance Settings) with a "语言 - Language" dropdown set to "中文 - Chinese simplified", a "主题" (Theme) dropdown set to "pmahomme", and a "字号" (Font Size) dropdown set to "82%"; "数据库服务器" (Database Server) showing details like "服务器: Localhost via UNIX socket", "服务器类型: MySQL", "服务器版本: 5.7.35 - MySQL Community Server (GPL)", "协议版本: 10", "用户: root@localhost", and "服务器字符集: UTF-8 Unicode (utf8)"; "网站服务器" (Web Server) showing details like "Apache/2.4.6 (CentOS) OpenSSL/1.0.2k-fips PHP/5.4.16 mod_perl/2.0.11 Perl/v5.16.3", "数据库客户端版本: libmysql - 5.6.51", and "PHP 扩展: mysqli"; and "phpMyAdmin" showing details like "版本信息: 4.0.10.20", "文档", "维基", "官方主页", "贡献", "获取支持", and "更新列表". A warning message at the bottom states: "phpMyAdmin 高级功能尚未完全设置，部分功能未激活。请点击[这里](#)查看原因。" (phpMyAdmin advanced features are not fully configured, some features are not activated. Please click [here](#) to view the reason).